

# COMP 3311: Database Management Systems

## Tutorial 7 Exercises MongoDB DBMS: MQL Aggregate Method

### What To Do

1. **Download** the zipped folder **T7Exercises.zip** from the *Tutorial 7* entry of the *Tutorial Assignments* section on Canvas to the desktop and unzip it. The folder contains the JSON files **customers.json** and **accounts.json** and the query files **t7Exercises1-4.txt** and **t7Exercises5-7.txt**.
2. **Connect** to your **MongoDB** account using **MongoDB Compass**.
3. **Create** a database named **bankdb** in your **MongoDB** account.
4. **Create** collections **customers** and **accounts** in your **bankdb** database and use the two JSON files **customers.json** and **accounts.json** to populate the **customers** and **accounts** collections, respectively.
5. **Modify** the **t7Exercises5-7.txt** query file by constructing **MQL** queries in the specified locations in the file.
6. **Test** your queries against the **bankdb** database to ensure that they execute without errors.

### What To Submit

Your modified **t7Exercises5-7.txt** file containing your MQL queries.

### How To Submit

**By 6 p.m. tomorrow**, upload your modified **t7Exercises5-7.txt** query file to Canvas.

**Note:** You will only be able to submit a file with extension txt.

### TEST YOUR QUERIES

The **MQL** queries in your submission will be tested by executing them against the **MongoDB bankdb** database. If any query raises an **MQL** error, then you will receive at most 0.5 marks for this exercise. Therefore, test your queries against the **bankdb** database before you submit them.

### bankdb database

#### customers collection

```
{_id: 1, name: "Amelia Earhart", account: "A1"},
{_id: 2, name: "Jody Foster", account: "A2"},
{_id: 3, name: "Fred Flintstone", account: "A3"},
{_id: 4, name: "Luke Skywalker", account: "A4"},
{_id: 5, name: "Conrad Black", account: "A5"}
```

#### accounts collection

```
{_id: "A1", balance: 10000,
  deposits: [{depositId: 1, customerId: 1, amount: 2000},
             {depositId: 2, customerId: 3, amount: 1000}],
  withdrawals: [{withdrawalId: 1, customerId: 1, amount: 1500},
                {withdrawalId: 2, customerId: 1, amount: 2000},
                {withdrawalId: 7, customerId: 3, amount: 500}],
_id: "A2", balance: 25000,
  deposits: [{depositId: 3, customerId: 1, amount: 1000},
             {depositId: 4, customerId: 2, amount: 3000},
             {depositId: 7, customerId: 2, amount: 1500}],
  withdrawals: [{withdrawalId: 3, customerId: 2, amount: 100},
                {withdrawalId: 4, customerId: 2, amount: 500},
                {withdrawalId: 5, customerId: 2, amount: 200}],
_id: "A3", balance: 20000,
  deposits: [{depositId: 5, customerId: 3, amount: 2000},
             {depositId: 6, customerId: 2, amount: 5000}],
  withdrawals: [{withdrawalId: 6, customerId: 3, amount: 1100}],
_id: "A4", balance: 15000,
  deposits: [{depositId: 8, customerId: 4, amount: 2500}],
_id: "A5", balance: 10000,
  withdrawals: [{withdrawalId: 8, customerId: 5, amount: 800}]}
```